Department of Engineering Cybernetics

TTK4265- Optical Remote Sensing

ASSIGNMENT 3

Data analysis with Satellite Ocean Color Data

Final report Length: max 10 pages

Deadline: 16 November 2026

Introduction and Background

In this assignment, you will work with real satellite data from two different sensors:

HYPSO-2, a hyperspectral satellite operated by NTNU.

Sentinel-3 OLCI, a multispectral satellite operated by ESA.

Each group will receive a pair of atmospherically corrected images from both satellites, covering the same region — either along the Norwegian coast, or in other coastal areas of Europe, or around the world.

The HYPSO-2 and Sentinel-3 OLCI data were atmospherically corrected using the OC-SMART model (http://www.rtatmocn.com/oc-smart/). The resulting images are called Level-2 products and are provided in HDF format.

The main goal of this assignment is to:

Understand the differences between hyperspectral and multispectral data, practice interpreting satellite-derived products such as remote sensing reflectance (Rrs), Chlorophyll-a (Chl-a), and Total Suspended Matter (TSM).

Tasks

Task 1: Plot the Remote Sensing Reflectance (Rrs) Spectra

Note: Before performing the task, please apply a CRS projection to both images.

Select at least four locations from the same region in both HYPSO-2 and Sentinel-3 OLCI images, and plot Rrs spectra from HYPSO-2 and Sentinel-3 OLCI side by side.

The locations should represent different water types — for example, from turbid coastal waters to clear open-ocean (blue) waters. RGB maps will be provided to help choose the locations, but only use the data cube for data extraction.

Extract the Rrs spectra (reflectance) for each selected point.

Create plots with:

X-axis: Wavelength (422–700 nm)

Y-axis: Rrs (in appropriate units)

Include a color legend that identifies each sensor and the information of the location.

Remember:

Sentinel-3 OLCI provides multispectral data with fewer and broader bands, whereas HYPSO provides high-resolution hyperspectral measurements (e.g., 422–750 nm). You need to plot the Rrs spectra from both sensors in the same figure. Each Rrs curve should have its own color and be clearly labeled in the legend as H2-Rrs and S3-Rrs.

For plotting, you can organize your data structure like this: first column is wavelength (422–700 nm), the second column is HYPSO hyperspectral Rrs, and the third column is multispectral Sentinel-3 OLCI Rrs.

Note: Remember, in the third column, you will have only a few Rrs values, because it is multispectral. Leave the other wavelengths for which you don't have OLCI Rrs as blank.

Since the HYPSO and Sentinel-3 OLCI central wavelengths do not match exactly, find each OLCI Rrs value and pair it with the nearest HYPSO wavelength within the 422–700 nm range.

This way, you can directly compare how both sensors measure the same regions' reflectance at hyperspectral and multispectral wavelengths. You only need to plot the Rrs data from 422 to 700 nm.

Finally, describe and interpret what you see in your plots.

How do the spectra differ between coastal and open-ocean waters?

Do you notice differences between HYPSO-2 and Sentinel-3 OLCI results for each location and bands?

Finally, provide the results like this: RGB image with the locations overlaid on the map, along with the reflectance comparison plot for each location.

Task 2: Map the Chlorophyll-a Concentration

Create Chlorophyll-a (Chl-a) maps from both HYPSO-2 and Sentinel-3 OLCI datasets.

Use the "chlor_a_oci_" from both sensors.

Display them side by side for comparison.

Use a common color scale (same minimum and maximum values) for both maps.

Include units of Chlorophyll-a on the color bar.

Briefly comment on the similarities and differences between the two maps.

Remember:

Sentinel-3 OLCI has a swath width of about 1500 km, while HYPSO-2 covers only around 300 km. When comparing the two datasets, you must match the spatial extent of Sentinel-3 to that of HYPSO-2 — in other words, extract or crop the Sentinel-3 Chlorophyll-a data so that it covers the same area as the HYPSO-2 scene.

Choose a chlorophyll-a concentration range that is appropriate for the region.

Task 3: Extract a Chl-a Gradient (Transect Analysis)

Draw a transect line across the image, extending from turbid coastal waters to clear open-ocean waters.

Extract Chlorophyll-a values along this transect for both satellites.

Use the "chlor_a_oci_" from both sensors.

Plot the extracted values:

X-axis: Latitude or longitude (depending on transect direction)

Y-axis: Chlorophyll-a concentration. Include the unit Chlorophyll-a.

Comment on the observed gradient — does Chlorophyll-a decrease as you move offshore? Are there differences between HYPSO-2 and OLCI?

Task 4: Map the Total Suspended Matter (TSM) Concentration

Generate TSM maps from both HYPSO-2 and Sentinel-3 OLCI datasets.

Use the "tsm_yoc_" from both sensors.

Display them side by side using a common color scale (same minimum and maximum values).

Include the unit of TSM.

Provide a short interpretation comparing the two sensors' outputs.

Remember:

Sentinel-3 OLCI has a swath width of about 1500 km, while HYPSO-2 covers only around 300 km. When comparing the two datasets, you must match the spatial extent of Sentinel-3 to that of HYPSO-2 — in other words, extract or crop the Sentinel-3 TSM data so that it covers the same area as the HYPSO-2 scene.

Task 5: Extract a TSM Gradient (Transect Analysis)

Using the same transect (or define a new one), extract TSM values from both satellites. The same as you did in Chlorophyll-a transect analysis.

Use the “tsm_yoc_” from both sensors.

Plot the extracted values:

X-axis: Latitude or longitude (depending on transect direction)

Y-axis: TSM concentration Include the unit of TSM.

Comment on how TSM changes from nearshore to offshore regions, and whether HYPSO-2 and OLCI show similar patterns.

General Notes

Make sure all your plots include axis labels, color bars, legends, and units.

Use consistent formatting so that visual comparisons are straightforward.

In your interpretation, focus on what you observe — not just describing colors or values, but explaining why these differences might occur (e.g., due to a different combination of water compositions, sensor characteristics, or spatial resolution).

Deliverables

Each group should submit:

A short report (max 5–6 pages) summarizing your results, including:

Figures

Explanations and interpretations

All plots (clearly labeled)

References